12/08/2026

Code-E

Aakash
Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18, Udyog Vihar, Gurugram, Haryana - 122015, ***Ph.***+91-1244168300

MM : 720 AIATS Prime for NEET-2027(XII Passed)_Unit Test-02 **Time : 180 Min.**

PHYSICS

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CHEMISTRY

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BOTANY

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ZOOLOGY

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Hints and Solutions

PHYSICS

(1) **Answer :** (1)
Solution:

$\vec{v}_{S/R} = \vec{v}_S - \vec{v}_R$

$\vec{v}_S = \vec{v}_{S/R} + \vec{v}_R$

In case of minimum time, angle between $\vec{v}_{S/R}$ and $\vec{v}_R$ is 90°.

$v_s = \sqrt{8^2 + 4^2}$

$= \sqrt{80}$

$\simeq 9$ m/s

(2) **Answer :** (2)
Solution:

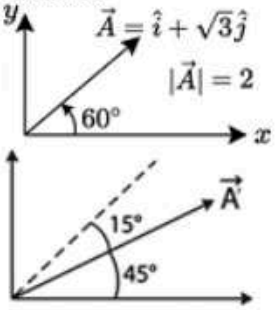
$\tan\alpha = \frac{5}{5} = 1 \Rightarrow \alpha = \frac{\pi}{4}$

(3) **Answer :** (4)
Solution:

$|\hat{i} + \hat{j}| = \sqrt{1^2 + 1^2} = \sqrt{2}$

Hence $\hat{i} + \hat{j}$ is not a unit vector.

(4) **Answer :** (3)
Solution:



$A_x = 2\cos 45^o$

$A_y = 2\sin 45^o$

New vector $\vec{A'} = \frac{2}{\sqrt{2}}\hat{i} + \frac{2}{\sqrt{2}}\hat{j} = \sqrt{2}\hat{i} + \sqrt{2}\hat{j}$.

$\frac{\vec{A'}}{|\vec{A'}|} = \hat{A} = \frac{\sqrt{2}\hat{i} + \sqrt{2}\hat{j}}{2} = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$.

(5) **Answer :** (2)
Solution:

$R = \sqrt{A^2 + A^2 + 2A^2\cos\theta}$

given $R = A\sqrt{3}$

⇒ θ = 60°

Resultant vector will be equally inclined at 30° from both the vectors.

(6) **Answer :** (2)
Solution:

$v_x = \frac{dx}{dt} = 6t^2$

$V_y = \frac{dy}{dt} = 8t$

$\vec{v} = 6t^2\hat{i} + 8t\hat{j}$

$t = 1\,\text{sec},\ \vec{v} = 6\hat{i} + 8\hat{j}$

$\Rightarrow |\vec{v}| = 10\ \text{m/s}$

(7) **Answer :** (1)

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Solution:

$\vec{v} = \vec{u} + \vec{a}t$

$= \left(4\hat{i} + 6\hat{j}\right) + \left(\hat{i} + \hat{j}\right) \times 2$

$= \left(6\hat{i} + 8\hat{j}\right)$ m/s

(8) **Answer :** (2)

Solution:

$t_1 = \frac{2u \sin\theta}{g}$

$t_2 = \frac{2u \cos\theta}{g}$

$t_1 \times t_2 = \frac{4u^2 \sin\theta \cos\theta}{g^2}$

$t_1 \times t_2 = \frac{2}{g}\frac{u^2 \sin 2\theta}{g}$

$t_1 \times t_2 = \frac{2}{g}R$

$t_1 \times t_2 \propto R$

(9) **Answer :** (2)

Solution:

$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B = \left(2\hat{i} + 3\hat{j}\right) - \left(\hat{i} + 2\hat{j}\right) = \left(\hat{i} + \hat{j}\right)$ m/s

(10) **Answer :** (1)

Hint:

Angular displacement $\theta = \omega t = x$

Solution:

$x_1 = \frac{2\pi}{60} \times 60 = \omega_1 \times 60$

$\frac{x_1}{\omega_1} = 60$

$x_2 = \frac{2\pi}{3600} \times 60 = \omega_2 \times 60$

$\frac{x_2}{\omega_2} = 60$

$\frac{x_1}{\omega_1} = \frac{x_2}{\omega_2}$

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(11) **Answer :** (2)

Hint:

$s_y = u_y t - \frac{1}{2}gt^2$

Solution:

Equation of trajectory

$u_y = u\sin\theta$

$t = \frac{x}{u\cos\theta}$

$\Rightarrow y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$

(12) **Answer :** (4)

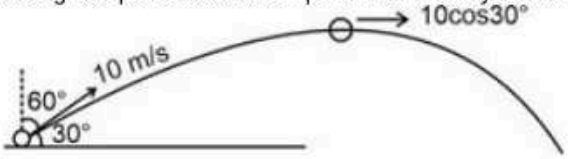
Solution:

For the ball's projectile motion from a height, its horizontal velocity remains same while its acceleration is equal to *g* (downwards).

(13) **Answer :** (4)

Solution:

At highest point vertical component of velocity become zero.



At highest point speed of object = 10cos30°

$= 5\sqrt{3}$ m/s

(14) **Answer :** (3)

Solution:

$\alpha = \frac{\omega_2 - \omega_1}{\Delta t} = \left(\frac{1200 - 600}{10}\right)\frac{2\pi}{60} = 2\pi$ rad/s^2

$\Rightarrow$ Use equation of motion, $\omega_2^2 = \omega_1^2 + 2\alpha\theta$

$\Rightarrow \omega_2 = 1200 \times \frac{2\pi}{60} = 40\pi$

$\omega_1 = 600 \times \frac{2\pi}{60} = 20\pi$

$(40\pi)^2 = (20\pi)^2 + 2 \times 2\pi\theta$

$\Rightarrow 1200\ \pi^2 = 4\pi\theta$

$\Rightarrow \theta = 300\pi$ radian

$\Rightarrow$ Number of revolution $= \frac{\theta}{2\pi} = 150$

(15) **Answer :** (1)
Solution:
We know, $R = 4H\cot\theta$
Here, $\theta = 45°$
$\therefore R = 4H\cot 45° \Rightarrow R = 4H$.

(16) **Answer :** (3)
Hint:
Range of projectile $R = \frac{2u_x u_y}{g}$
Solution:
$R = \frac{2\times 10\times 20}{10} = 40$ m

(17) **Answer :** (4)
Solution:
$x = R$, $y = 0$
$0 = 12R - \frac{3}{4}R^2$
$R = 16$ m

(18) **Answer :** (4)
Solution:
The angle is 90° for uniform circular motion only. For circular motions with increasing and decreasing speeds, the angle is acute and obtuse respectively.

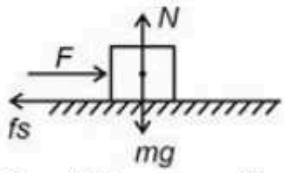
(19) **Answer :** (1)
Solution:
$\tan\theta = 2$ and $u_x = u\cos\theta = 5$ m/s
$u_y = u\sin\theta = u\cos\theta \times \tan\theta = 5 \times 2 = 10$ m/s
a(ii) $T = \frac{2u_y}{g} = \frac{2\times 10}{10} = 2$ s
c(iii) $H = \frac{u_y^2}{2g} = \frac{10^2}{2\times 10} = 5$ m
b(i) $R = \frac{2u_x u_y}{g} = \frac{2\times 5\times 10}{10} = 10$ m

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(20) **Answer :** (4)
Hint:
Time of flight = $t_1 + t_2$
Solution:
Time of flight = 3 + 5 = 8 s
$\frac{2u\sin(\theta)}{g} = 8$
$\frac{2\times u\times\sin(30^\circ)}{10} = 8$
$u = 80$ m/s

(21) **Answer :** (1)
Solution:
$\vec{v} = \vec{v_y} + \vec{v_x} = v_y\hat{j} + v_x\hat{j}$
$v_y = \sqrt{2gh}$
$= \sqrt{2\times 9.8\times 4.9}$
$= 9.8\ \text{m s}^{-1}$
$v_x = 9.8\ \text{m s}^{-1}$
$\tan\theta = \frac{v_y}{v_x} = \frac{9.8}{9.8} = 1$
$\theta = \tan^{-1}(1) = 45°$

(22) **Answer :** (2)
Hint:
Block will move only if $F > (f_S)_{max}$
Solution:



Here, $(f_S)_{max} = \mu_S \times N$
$= 0.4 \times 50$
$= 20$ N
$F < (f_S)_{max}$, so block will not move.
Hence, friction on block, $f = F = 16$ N

(23) **Answer :** (1)
Solution:

$y = x - \frac{x^2}{4}$

$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$

$\tan\theta = 1$

$\theta = 45°$

$\frac{g}{2u^2\cos^2\theta} = \frac{1}{4}$

$\frac{2u^2}{2} = 40$

$u = 2\sqrt{10}\ \text{ms}^{-1}$

(24) **Answer :** (4)
Solution:

$m = \frac{F}{a} = \frac{\sqrt{6^2+8^2+10^2}}{1} = \sqrt{200} = 10\sqrt{2}\,\text{kg}$

(25) **Answer :** (2)
Hint:

$a = \frac{(m_2-m_1)}{(m_2+m_1)}g$

Solution:

$a = \frac{(2-1)}{(2+1)}g = \frac{10}{3}\,\text{m/s}^2$ for both blocks.

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(26) **Answer :** (1)
Solution:

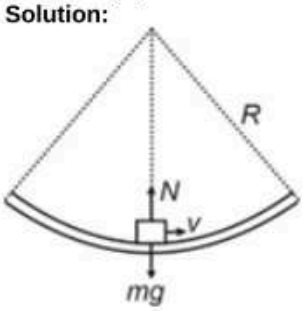
$F = \left(\frac{\Delta p}{\Delta t}\right) = \frac{n(mu-0)}{(1)} = mnu$

(27) **Answer :** (3)
Solution:

$V = \frac{mu}{M}$

$V = \frac{0.1}{2} \times (200) = 10$ m/s

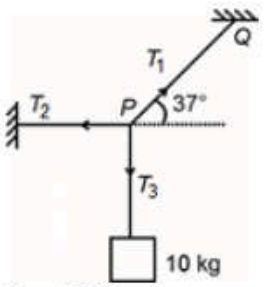
(28) **Answer :** (2)
Solution:



$N - mg = \frac{mv^2}{R}$

$N = mg + \frac{mv^2}{R}$

(29) **Answer :** (4)
Solution:



$T_1 \sin 37° = mg$

$T_1 \times \frac{3}{5} = 100$

$\Rightarrow \ T_1 = \frac{500}{3}$N

(30) Answer : (4)
Solution:
All the statements are true
as $\vec{F} = \frac{d\vec{p}}{dt}$

$\vec{F} = m\vec{a}$

(31) Answer : (3)
Solution:
As downward acceleration of elevator $a > g$, block loses the contact with the elevator and its acceleration is g downward.
So speed of block at $t = 1$ s,
$v = u + gt = 10 \times 1 = 10$ m/s

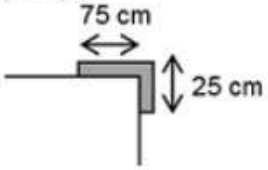
(32) Answer : (1)
Solution:
$F = x^2 - 6x + 8$
For equilibrium, $F = 0$
So, $x^2 - 6x + 8 = 0$
$x^2 - 4x - 2x + 8 = 0$
$x(x-4) - 2(x-4) = 0$
$x = 2$ m and 4 m
distance $= x_2 - x_1 = 2$ m

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(33) Answer : (2)
Solution:
Initially friction will be equal to the applied force. Later it becomes constant (kinetic friction).

(34) Answer : (3)
Solution:
$v_m = \sqrt{\mu \text{Rg}}$
$= \sqrt{0.2 \times 32 \times 10} = 8$ m/s

(35) Answer : (1)
Solution:
$\frac{M}{4}g = \frac{\mu 3M}{4}g$
$\mu = 1/3$



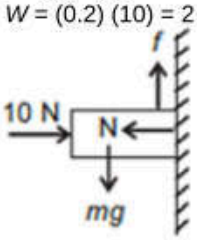
(36) Answer : (2)
Solution:
$T_1 = (m_B + m_C)\, a$
$T_2 = (m_C)\, a$
$\frac{T_1}{T_2} = \frac{(m+3m)}{3m} = \frac{4}{3}$

(37) Answer : (4)
Solution:
$mg - T = ma$
$mg - \frac{2mg}{3} = ma$

$a = \frac{g}{3}$

(38) **Answer :** (1)
Solution:
When standing at rest reading = 60 kg
$\therefore m$ = 60 kg
When he jumps upward, reading = 70 kg
So weighing machine measures normal reaction
$(R - mg) = ma$
$(70 - 60)g = 60a$
$a = \frac{g}{6}$

(39) **Answer :** (4)
Solution:
Normal reaction N = 10 N
In vertical direction, frictional force will balance its weight
$f = mg = \mu N$
$W = (0.2)(10) = 2$ N



(40) **Answer :** (4)
Solution:
$a = \frac{F}{m_1 + m_2} = \frac{40}{5} = 8$ m/s^2
$40 - kx = 2 \times 8$
$x = \frac{24}{400} = 6$ cm

(41) **Answer :** (1)
Solution:
$F = m\omega^2 r$
$F' = (m)(2\omega)^2\left(\frac{r}{2}\right) = 2m\omega^2 r$

(42) **Answer :** (1)
Solution:
Momentum = $\int F dt$ = area under $F - t$ curve
$= \frac{10 \times 1}{2} + 10 \times 2$
= 25 Ns

(43) **Answer :** (3)
Solution:
The reading of spring balance will be equal to $mg\sin\theta$.
$\Rightarrow 10 \times 10 \times \frac{1}{2} = 50$ N

(44) **Answer :** (2)
Solution:
$\tan\theta = \frac{v^2}{rg} = \frac{10 \times 10}{10\sqrt{3} \times 10} = \frac{1}{\sqrt{3}}$
$\Rightarrow \theta = 30°$

(45) **Answer :** (1)
Solution:
Under Newton's 3rd law, action and reaction are always equal and opposite. They always act on different bodies.

CHEMISTRY

(46) **Answer :** (4)
Solution:
Law of octaves: When elements are arranged in increasing order of their atomic weights, every eighth element has properties similar to the first element. This law seemed to be true only for elements up to calcium.

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(47) **Answer :** (2)
Solution:

	Element	Electronegativity value on pauling scale
1.	Be	1.5
2.	B	2.0
3.	C	2.5
4.	S	2.5
5.	Al	1.5
6.	P	2.1
7.	Na	0.9
8.	Mg	1.2

(48) **Answer :** (4)
Solution:
The oxidation state of Al is +3 and covalency is 6.
$x - 1 + 0 \times 5 = +2$
$x = +3$

(49) **Answer :** (3)
Solution:
Given electronic configuration is of Ga(31).
The atomic number of the element just above it is 31 – 18 = 13

(50) **Answer :** (3)
Solution:
Tellurium is a chalcogen (*p*-block element).

(51) **Answer :** (3)
Solution:
Elements of *s* and *p* blocks are called representative elements.

(52) **Answer :** (2)
Hint:

Electronic configuration	Element
$1s^2 2s^2$	Be
$1s^2 2s^2 2p^1$	B
$1s^2 2s^2 2p^3$	N
$1s^2 2s^2 2p^4$	O

Solution:
Due to stable electronic configuration of 'Be' & 'N', they have higher first ionization enthalpy than B and O respectively
B < Be < C < O < N < F

(801 889 1086 1314 1402 1681) $\Delta_i H / kJ\,mol^{-1}$

(53) **Answer :** (3)
Solution:
Be is in diagonal relationship with Al

(54) **Answer :** (2)
Solution:
Size of anion is more than parent atom due to less effective nuclear charge on outermost electrons. Size of cation is less than parent atom due to more effective nuclear charge.
$\therefore I^- > I > I^+$ size order.

(55) **Answer :** (2)
Solution:
Ga is known as eka-aluminium.

(56) **Answer :** (2)
Solution:
Z = 105 : Unnilpentium (Unp).

(57) **Answer :** (1)
Solution:
Chlorine has highest negative electrons gain enthalpy which belong to 3rd period group 17 in periodic table.

(58) **Answer :** (1)

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Solution:

Element	Electronegativity value
Si	1.8
C	2.5
N	3.0
O	3.5

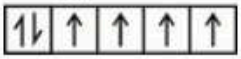
(59) **Answer :** (4)
Solution:

Ion	No. of electrons
Nickel	Transition metal
Chlorine	Non-metal
Germanium	Metalloid
Samarium	Lanthanoid

(60) **Answer :** (2)
Hint:
Element of *s* and *p* block are collectively called representative elements.
Solution:
Bromine is present in group 17 and period 3 while sulphur is in group 16 and period 3.

Element	Electronegativity (on Pauling scale)
S	2.5
Br	2.8

- Barium– *s* block element (Alkaline earth metal)
- Boron – *p* block element (Group 13)
- Both Sn and Pb are the members of group 14
- $_{26}Fe = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6\ 4s^2$



4 unpaired electrons

(61) **Answer :** (1)
Hint:
Dubnium has atomic number 105 and has IUPAC symbol Db.
Solution:

	IUPAC official name and IUPAC symbol	Atomic number	Name according to IUPAC nomenclature
a.	Dubnium (Db)	105	Unnilpentium
b.	Seaborgium (Sg)	106	Unnilhexium
c.	Bohrium (Bh)	107	Unnilseptium
d.	Hassium (Hs)	108	Unniloctium

(62) **Answer :** (4)
Hint:
When an electron is added to a neutral gaseous atom to convert it into a negative ion, the enthalpy change accompanying the process is defined as the electron gain enthalpy.
Solution:
Correct order of negative electron gain enthalpy is O < Te < Se < S.

(63) **Answer :** (3)
Solution:
According to Dobereiner's traid, the middle element of each of the triads had an atomic weights about half way between the atomic weights of the other two elements. Dobereiner's traid has similar physical and chemical properties.

(64) **Answer :** (4)
Solution:
The metallic character increases down the group and decreases along a period as we move from left to right.

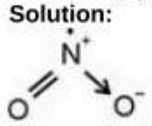
(65) **Answer :** (2)
Hint:
Bond length $\propto \frac{1}{\text{Bond order}}$
Solution:

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$$\text{Bond order :} \quad \underset{3}{\underset{\text{(Maximum)}}{CO}} > \underset{2}{CO_2} = C_3O_2 > \underset{1.33}{\underset{\text{(Minimum)}}{CO_3^{2-}}}$$

Therefore, maximum bond length of C – O is in CO_3^{2-}.

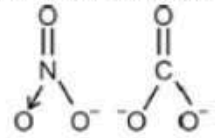
(66) **Answer :** (3)
Solution:



It is an odd electron molecule.

(67) **Answer :** (3)
Solution:

Number of electrons in NO_3^- as well as CO_3^{2-} = 32

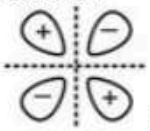


Trigonal planar geometry

(68) **Answer :** (2)
Solution:

Paramagnetic- NO, N_2^+ and O_2^-

(69) **Answer :** (2)
Solution:

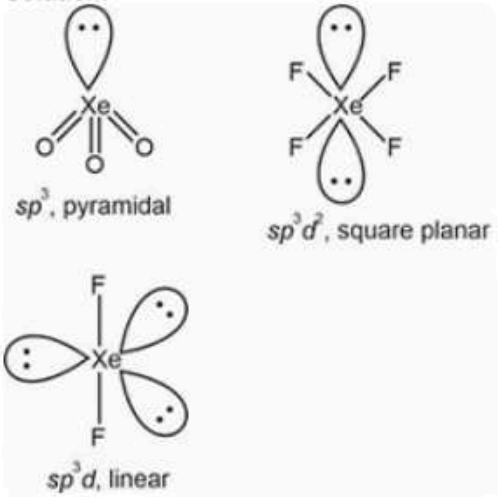


represents π-antibonding molecular orbital.

(70) **Answer :** (3)
Solution:
Number of hybrid orbitals are equal to the number of atomic orbitals involved in hybridisation.

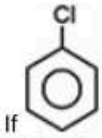
(71) **Answer :** (1)
Solution:
An atomic orbital is monocentric as it is influenced by one nucleus only and molecular orbital is polycentric as it is being influenced by 2 or more nuclei.

(72) **Answer :** (4)
Solution:



(73) **Answer :** (2)
Solution:

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If has x dipole moment, (i) and (iii) has x dipole moment, (ii) has 2x dipole moment, (iv) and (v) has zero dipole moment.

(74) **Answer :** (3)
Solution:
Boiling point of HF is higher than HCl due to intermolecular hydrogen bond.

(75) **Answer :** (1)
Solution:
In $N(SiH_3)_3$ p π – d π back bonds are formed.

(76) **Answer :** (1)
Solution:
N_2 and NO^+ are isoelectronic species and have bond order = 3

(77) **Answer :** (4)
Solution:

Species	Bond Order
O_2^-	1.5
O_2	2
O_2^+	2.5

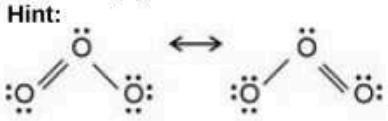
Lesser is the bond order, longer is the bond length.

(78) **Answer :** (1)
Solution:

Bond type	Covalent bond length (pm)
C – H	107
C – N	143
C – C	154

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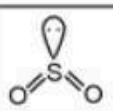
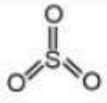
(79) **Answer :** (4)
Hint:



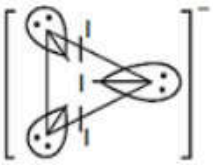
2-canonical forms of ozone

Solution:
Formal charge on central oxygen atom in O_3 formal charge $= 6 - \frac{1}{2} \times 6 - 2 = +1$
Shape of ozone is bent.

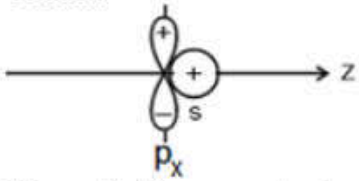
(80) **Answer :** (4)
Solution:

	σ bonds	π bonds
O=C=O	2	2
	2	2
	3	3
H–C≡C–H	3	2

(81) **Answer :** (3)
Solution:
Hybridisation of central I atom is sp^3d and three lone pairs are present at central atom.



(82) **Answer :** (1)
Solution:



This results into zero overlapping

(83) **Answer :** (1)
Solution:

Na^+ has larger size and less charge as compared to Al^{3+} and hence has more ionic nature

Cs^+ is larger in size hence it forms more ionic compound with I^- ion as compared to Na^+

Al^{3+} is smaller with more charge and hence it forms more covalent compound than Mg^{2+}

(84) **Answer :** (2)
Hint:
The lone pair (lp) of electrons in a molecule occupy more space as compared to bond pair (bp) of electrons.
Solution:
There is greater repulsion between lone pair of electrons as compared to lone pair-bond pair and bond pair – bond pair repulsions.
The correct order of repulsive interactions is lp – lp > lp – bp > bp – bp

(85) **Answer :** (2)
Hint:

• Percentage ionic character = $\frac{\mu_{observed}}{\mu_{calculated}} \times 100$

• 1D = 3.33×10^{-30} Cm
Solution:

$\mu_{observed} = 0.8 \times 3.33 \times 10^{-30}$ Cm

$\mu_{calculated} = 1.602. \times 10^{-19} \times 100 \times 10^{-12}$ Cm

$= 1.6 \times 10^{-29}$ Cm

Percentage ionic character = $\frac{0.8 \times 3.33 \times 10^{-30} \times 100}{1.6 \times 10^{-29}}$

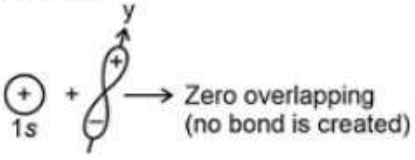
$= 1.665 \times 10 = 16.65\%$

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(86) **Answer :** (3)
Solution:

BF_3	120°
XeF_4	180° and 90°
PCl_5	120°, 90°, 180°
CO_2	180°

(87) **Answer :** (4)
Solution:



(88) **Answer :** (4)
Solution:

$$F_2 : \sigma 1s^2,\ \sigma^* 1s^2,\ \sigma 2s^2,\ \sigma^* 2s^2,$$
$$\sigma 2p_z^2,\ \pi 2p_x^2 = \pi 2p_y^2,\ \pi^* 2p_x^2 = \pi^* 2p_y^2$$

Highest occupied molecular orbital is π^*2p_x or π^*2p_y.

(89) **Answer :** (4)

Solution:
For He_2 molecule

Electronic configuration is $\sigma 1\ s^2$, $\sigma^* 1\ s^2$
so bond order = $\frac{1}{2}[N_b - N_a]$

$= \frac{1}{2}[2-2]$
$= 0$

Since, bond order is zero, so He_2 molecule does not exist.

(90) **Answer :** (4)
Solution:
Correct order of lattice enthalpy
LiF > NaF > LiCl > NaCl

BOTANY

(91) **Answer :** (3)
Solution:
Standard rules for Binomial nomenclature were established by Carolus Linnaeus.

(92) **Answer :** (3)
Solution:
Plant family solanaceae are included in the order polymoniales mainly based on the floral characters.

(93) **Answer :** (2)
Hint:
The taxonomic hierarchy is as follows:

Kingdom
↑
Phylum or Division
↑
Class
↑
Order
↑
Family
↑
Genus
↑
Species

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Solution:
The correct depiction of sequential order of these taxa in ascending order is as follows:
Poaceae → Poales → Monocotyledonae → Angiospermae → Plantae

(94) **Answer :** (3)
Solution:
1. Diptera, sapindales and poales are orders in the taxonomic hierarchy.
2. Other options have taxa of different taxonomic categories.

(95) **Answer :** (4)
Solution:
Systematics takes into account evolutionary relationships between organisms.
The word systematics is derived from the Latin word 'Systema' which means systematic arrangement of organisms.

(96) **Answer :** (1)
Solution:
Housefly belongs to the family Muscidae. Mango belongs to the family Anacardiaceae.

(97) **Answer :** (2)
Solution:
Classification of organisms into various categories help in identifying similarities and dissimilarities among different kinds of organism.

(98) **Answer :** (2)
Solution:
The number of similar characteristics goes on decreasing from species to kingdom.

(99) **Answer :** (2)
Solution:

There are about 1.7-1.8 million species according to IUCN which are known and described.

(100) Answer : (3)
Solution:
Diptera and Insecta are order and class, respectively.

(101) Answer : (2)
Solution:
All the three words (generic name, specific epithet and authors citation) collectively form binomial epithet.
The correct binomial epithet of mango is *Mangifera indica* Linn.

(102) Answer : (4)
Solution:
Dinoflagellates are photosynthetic protists and have chlorophyll pigments.

(103) Answer : (4)
Solution:
Diptera, Primata and Sapindales are orders. Order comes between class and family.

(104) Answer : (2)
Solution:
Taxonomy is a process of classification that is based on the characteristics of all living organisms that can be classified into different taxa.
Systematics takes into account evolutionary relationships between organisms.
Classification is a process by which anything is grouped into convenient categories based on some easily observable characters.
Taxonomic categories are distinct biological entities and not merely morphological aggregates.

(105) Answer : (2)
Solution:
The ascending order of taxonomic hierarchy is
Species → Genus → Family → Order → Class → Division → Kingdom

(106) Answer : (2)
Solution:
Mango and potato belong to orders Sapindales and Polymoniales respectively but they belong to the same class Dicotyledonae.

(107) Answer : (4)
Solution:
Protozoans lack cell wall.

(108) Answer : (1)
Solution:
Blue green algae have chlorophyll a similar to green plants.

(109) Answer : (2)
Solution:

Coccus	–	Spherical
Bacillus	–	Rod-shaped
Spirillum	–	Spiral
Vibrium	–	Comma-shaped

(110) Answer : (2)
Hint:
Cilia and cell wall are not found in euglenoids.
Solution:
Euglenoids have a covering of pellicle which is proteinaceous. This covering provides flexibility to the organisms.

(111) Answer : (3)
Solution:
Penicillium – Ascomycetes member with sexual spores called ascospores.
Rhizopus – Zygomycetes member with asexual spores called sporangiospores.
Ustilago – In the member of Basidiomycetes asexual spore are generally not found.
Alternaria– Some members of class Deuteromycetes are saprophytes or parasites.

(112) Answer : (4)
Solution:
Cuscuta is a parasitic plant with heterotrophic mode of nutrition.

(113) Answer : (3)
Solution:
Viroids caused potato spindle tuber disease.

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(114) Answer : (3)
Solution:
Sexual spore in Basidiomycetes is basidiospore.

(115) Answer : (4)
Solution:
Organisation at tissue level is absent in Protista as they are unicellular eukaryotes.

(116) Answer : (1)
Solution:
Examples of symbiotic N_2-fixing bacteria are *Rhizobium, Frankia, Xanthomonas.*
Examples of free-living N_2-fixing bacteria are *Azotobacter, Clostridium, Beijerinckia.*

(117) Answer : (3)
Solution:
Fungi prefer to grow in warm and moist places.

(118) Answer : (1)
Solution:
Karyogamy is the fusion of nuclei that occurs after plasmogamy. In some members of Phycomycetes, zygospores are formed after karyogamy. These zygospores are the site of meiosis.

(119) Answer : (4)
Solution:
Ustilago - Basidiospores, *Claviceps* - Ascospores, *Mucor* - Zygospores, *Trichoderma* - Conidia

(120) Answer : (1)
Solution:

Morels	Edible fruiting body
Puccinia	Rust fungus
Alternaria	Only vegetative phase is known
Albugo	Aseptate and coenocytic mycelium

(121) Answer : (4)
Solution:
Viroids are sub-viral agents as infectious RNA particles, without protein coat.

(122) Answer : (4)
Hint:
Spirogyra is not a unicellular organism.
Solution:
Kingdom Protista has brought together *Chlamydomonas, Chlorella* with *Paramoecium* and *Amoeba*.

(123) Answer : (4)
Solution:
Creutzfeldt-Jacob disease in humans is caused by prions.

(124) Answer : (2)
Solution:
Zygospores are diploid sexual spores formed by members of Phycomycetes.

(125) Answer : (1)
Hint: Three-domain system divides the kingdom Monera into two domains.
Solution:
Answer (1)
Three domains are Bacteria, Archaea and Eukarya. All the eukaryotes are grouped in Eukarya.

(126) Answer : (3)
Solution:
In slime moulds, spores have cellulosic cell wall similar to plants and main body without cell wall like animals.

(127) Answer : (3)
Solution:
The network of hyphae is known as mycelium.

(128) Answer : (2)
Solution:
Basidiospores are sexual haploid spores produced on basidium.

(129) Answer : (4)
Solution:
Slime moulds produce plasmodium under favourable condition.

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(130) Answer : (3)
Hint:
The given organism is a dividing bacterium
Solution:
'X' represents the non-cellulosic cell wall.
'Y' represents the cell membrane.
'Z' represents the circular DNA.

(131) Answer : (3)
Solution:
Members of Deuteromycetes lack sexual reproduction.

(132) Answer : (3)
Solution:
Methanogens are methane producing bacteria abundant in marshy area and gut of ruminants.

(133) Answer : (4)
Solution:
Some of the bacteria are autotrophic but vast majority are heterotrophic.

(134) Answer : (2)
Hint:
Lichens are symbiotic association between algae and fungi.
Solution:
Algal component is known as phycobiont and fungal component as mycobiont.

(135) Answer : (3)
Solution:
For the first time, W.M. Stanley showed that viruses could be crystallised.

ZOOLOGY

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(136) Answer : (1)
Solution:
CO_2 is carried by haemoglobin as carbamino-haemoglobin (about 20-25 percent).

(137) Answer : (1)
Solution:
Diaphragm contracts to increase the thoracic volume in antero-posterior axis.

(138) Answer : (4)
Hint:
ERV + TV + IRV
Solution:
Vital capacity (VC) is defined as the maximum volume of air a person can breathe in after a forceful expiration or the maximum volume of air a person can breathe out after a forceful inspiration.
Depending upon the age, sex and height of an individual, its value varies from 4000 mL to 4600 mL.
Vital capacity (VC) = ERV + TV + IRV or IC + ERV = 4000 mL to 4600 mL.

(139) Answer : (1)
Hint:
Normal blood pressure is 120/80 mmHg.
Solution:
Hypertension or repeated high blood pressure is a condition in which pressure of blood is raised in arteries. High blood pressure of 200/100 mmHg may harm various organs of the body such as kidneys and brain to a great extent.

(140) Answer : (2)
Solution:
SA node generates maximum number of action potentials.

(141) Answer : (3)
Solution:
By counting the number of QRS-complexes that occur in a given time period, one can determine the heart rate of an individual.

(142) Answer : (3)
Hint:
Composition of plasma
Solution:
Lymph has the same mineral distribution as that in plasma. Exchange of nutrients, gases, *etc.*, between blood and the cells always occur through this fluid. Lymph is a colourless fluid containing specialised lymphocytes which are

responsible for immune responses of the body. Lymph is also an important carrier for nutrients, hormones, *etc*. Fats are absorbed through lymph in the lacteals present in the intestinal villi.

(143) Answer : (2)

Hint:

CO_2 is end product of metabolism in tissues.

Solution:

If a person holds his breathe *i.e*., fresh air is not inhaled, then fresh supply of O_2 is arrested. Concentration of CO_2 however continues to rise in blood producing an urge to breathe.

(144) Answer : (1)

Solution:

Basophils secrete histamine, serotonin and heparin and lymphocytes constitute 20-25% of total WBCs. T-lymphocytes mature in thymus gland.

(145) Answer : (2)

Solution:

Parasympathetic nerve endings release acetylcholine which decreases rate of heart beat.

(146) Answer : (1)

Solution:

Total lung capacity (TLC) is the total volume of air accomodated in the lungs at the end of forced inspiration.
Inspiratory capacity is the total volume of air inspired after a normal expiration. Thus, it includes TV (Tidal volume) and IRV (Inspiratory reserve volume).
Vital capacity is the maximum volume of air a person can breathe in after a forceful expiration or maximum volume of air a person can expire after a forceful inspiration.
FRC is the volume of air that remains in the lungs after normal expiration.

(147) Answer : (1)

Hint:

Erythroblastosis foetalis.

Solution:

Rh incompatibility occurs due to crossing over of Anti-D antibodies (IgG) from maternal to foetal circulation across the placenta. This occurs when mother is Rh –ve and the foetus in Rh +ve.

(148) Answer : (4)

Solution:

Chordae tendineae prevent AV valves from prolapsing into atria during ventricular contraction.

(149) Answer : (1)

Solution:

Rise in cardiac output helps by enhancing the supply of nutrients and oxygen to the contracting muscles.
The athlete has to perform more work compared to that of an ordinary man. For performing those physical activities, more blood flow is required.

(150) Answer : (2)

Solution:

At high altitude, atmospheric O_2 level is less and hence, more RBCs are needed to absorb the required amount of O_2 to survive.

(151) Answer : (2)

Solution:

Various substances dissolved in plasma are transported to different tissues of the body.

(152) Answer : (1)

Solution:

Receptors associated with aortic arch and carotid artery recognize changes in CO_2 and H^+ concentration.

(153) Answer : (2)

Hint:

AVN is the pacesetter of heart.

Solution:

SAN has the highest inherent capacity of impulse generation, *i.e*., it generates highest, number of action potentials per minute.

(154) Answer : (3)

Solution:

The lungs are situated in the thoracic chamber which is anatomically an air-tight chamber.

(155) Answer : (4)

Solution:

The trachea, primary, secondary and tertiary bronchi and initial bronchioles are supported by incomplete cartilaginous rings.

(156) Answer : (4)

Hint:

It is the sum of ERV and RV

Solution:

FRC = ERV + RV = 2100-2300 mL.

(157) Answer : (4)

Solution:

Branchial respiration or respiration *via* gills is seen in aquatic arthropods and molluscs. Cutaneous refers to skin and pulmonary to lungs.

(158) Answer : (1)

Solution:

A specialised centre present in the medulla region of the brain called respiratory rhythm centre is primarily responsible for this regulation. Another centre present in the pons region of the brain called pneumotaxic centre can moderate the functions of the respiratory rhythm centre. Neural signal from this centre can reduce the duration of inspiration and thereby alter the respiratory rate.

(159) Answer : (2)

Solution:

A spirometer is a device used to measure the volume of air inhaled and exhaled by the lungs. However, it cannot measure certain lung volumes directly. Residual Volume (RV) is the volume of air remaining in the lungs after a maximal exhalation, and it cannot be measured by a spirometer because it is the air that remains in the lungs and airways after the most forceful expiration.

(160) Answer : (1)

Hint:

Universal blood donor.

Solution:

Blood Groups and Donor Compatibility

Blood Group	Antigens on RBCs	Antibodies in Plasma	Donor's Group
A	A	anti-B	A, O
B	B	anti-A	B, O
AB	A, B	nil	AB, A, B, O
O	nil	anti-A, B	O

(161) Answer : (4)

Solution:

Angina pectoris refers to acute chest pain which occurs as a result of less oxygen reaching the heart muscles.

(162) Answer : (2)

Solution:

Every 100 mL of oxygenated blood can deliver around 5 mL of O_2 to the tissues under normal physiological conditions.

Then 2L oxygenated blood can deliver around 100 mL of O_2 to the tissues under normal physiological conditions.

(163) Answer : (3)

Solution:

Plasma without clotting factors is called serum.

(164) Answer : (2)

Hint:

The flow of blood in veins is slower than in arteries.

Solution:

Veins have valves which prevent backward flow of blood.

(165) Answer : (3)

Hint:

Function of exchange part.

Solution:

Conducting part of human respiratory system transports the atmospheric air to the alveoli, clears it from foreign particles, humidifies and also brings the air to body temperature.

Exchange part is the site of actual diffusion of O_2 and CO_2 between blood and atmospheric air.

(166) Answer : (2)

Solution:

Systemic arteries have oxygenated blood having pO_2 95 mm Hg and pCO_2 40 mm Hg.

Respiratory Gas	Atmospheric Air	Alveoli	Blood (Deoxygenated)	Blood (Oxygenated)	Tissues
O_2	159	104	40	95	40

CO_2	0.3	40	45	40	45

(167) Answer : (2)
Solution:
Cardiac arrest is a condition in which the heart stops beating.

(168) Answer : (4)
Solution:
Partial pressure of oxygen in oxygenated blood is 95 mmHg and in deoxygenated blood, it is 40 mmHg.

(169) Answer : (2)
Solution:
Renal portal system is present in fishes and amphibians, reduced in reptiles and birds and absent in mammals.

(170) Answer : (4)
Hint:
Thrombokinase catalyses conversion of prothrombin to thrombin.
Solution:

$$\text{Fibrinogen} \xrightarrow{\text{Thrombin}} \text{Fibrin}$$

(171) Answer : (1)
Solution:
Electrocardiogram is the graphical representation.

(172) Answer : (4)
Solution:
The first step at the start of respiration is breathing or inspiration. Followed by exchange of gases between deoxygenated blood and alveoli.

(173) Answer : (2)
Solution:
Breathing rate in an adult human under normal conditions is 12-16 times per minute and is nearly 40 times per minute in newly born.

(174) Answer : (2)
Solution:
Medulla oblongata is a vital part and centre for respiration.

(175) Answer : (1)
Solution:
Closure of semilunar valves occurs before opening of cuspid valves during ventricular diastole.

(176) Answer : (2)
Solution:
During expiration, relaxation of muscles brings the ribs and sternum back to normal position which reduces the volume of thoracic chamber. This leads to increase in intra-pulmonary pressure - positive pressure in lungs with respect to atmospheric pressure.

(177) Answer : (2)
Solution:
Heart sound 'dub' is produced by closing of semilunar valves at the beginning of ventricular diastole.

(178) Answer : (4)
Hint:
Think about basement membrane
Solution:
At the tissue site, where pCO_2 is high due to catabolism, CO_2 diffuses into RBCs and forms HCO_3^- and H^+ in the presence of enzyme carbonic anhydrase. Diffusion membrane is formed by two cellular layers and an acellular basement substance in between them.

(179) Answer : (2)
Hint:
Phase of cardiac cycle in which volume of heart chambers increases.
Solution:
Blood from pulmonary veins and vena cava fills the left and right atria, respectively. The pressure causes AV valves to open and blood to flow from atria into ventricles.

(180) Answer : (3)
Solution:
The valves in the heart and blood vessels allow the flow of blood only in one direction. These valves prevent any backward flow of blood.

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